# Virtual Reality Life-Saving

Basic Life Support & Defibrillation Training

**Maurizio Meschi, Nicolò Trebino**



UniGe | DIBRIS

# 01. INTRODUCTION

# 02. WHAT WE LEARNED

# 03. CHALLENGES AND PROBLEMS

# 04. FUTURE IMPLEMENTATIONS

# 05. DEMO

# 01. Introduction

*Empowering everyone through immersive simulation and AI-guided training*

# Project Overview

VR-BLSD is an immersive simulation designed to teach the fundamentals of **Basic Life Support and Defibrillation (BLSD)**.

Developed in **Unreal Engine 5** for Meta Quest, it allows users to practice in high-fidelity emergency scenarios, reducing the barrier to action in real-world cardiac arrest situations.

## Motivation

While not a substitute for official certification, VR-BLSD acts as a vital first approach to raise public awareness, overcome hesitation, and introduce everyone to essential life-saving skills.

# Innovative AI Tutor

## Real-time Conversational Support

Powered by **MetaHuman** and **ConvAI**, the virtual instructor Hadley acts as an emotional anchor.

She provides adaptive procedural feedback, guiding the user through stress-inducing scenarios while maintaining authoritative yet empathetic mentorship.

# 02. What We Learned

*Technical Foundations and Psychological Integration*

# Technical Architecture & Logic

## Modular Design

Implemented parent classes to ensure scalable and reusable interaction logic across all BLSD equipment.

## Optimization

Mastered performance tuning for mobile VR, balancing high-fidelity MetaHumans with baked lighting and optimized textures.

## Cross-Platform

Engineered a unified code structure that supports both immersive VR headsets and traditional Desktop input methods.

# 03. Challenges & Problems

*Technical Foundations and Psychological Integration*

# Hardware and Logistical Barriers

**Resource Allocation**: Limited to a single high-end workstation, we adopted pair-programming to maintain a unified and consistent logic flow.

**Hardware Access**: Restricted lab hours for Meta Quest testing were managed by a Desktop-first development strategy for core logic.

**Iterative Testing**: Synchronizing physical lab time with software milestones required meticulous project scheduling.

# Beta Software Volatility

Developing with the **ConvAI beta** plugin introduced frequent breaking changes. We navigated this by:

- Locking the plugin to a **stable version** during critical sprints.
- Implementing a **100% custom Push-to-Talk** system to resolve native plugin instability.
- Implementing a **custom user tracking** to increase presence and realism of the conversation.
- Implementing ConvAI agent **proactive dialogue**
- Implementing custom agent **spatial awareness** and autonomous **path finding**

Mar 9
github-actions
4.0.0-beta.12
2625c66
Compare

**Convai Unreal Engine SDK Version 4.0.0-beta.12**

- Deprecated old ConvaiBaseCharacter and ConvaiBasePlayer BP Classes.
- Added LLM start/stop events (OnLLMStarted, OnLLMStopped) to ConvaiChatbotComponent.
- Added emotions support.
- Added toggle STT functionality.
- Improved lip sync at the end of response.
- Fixed lip sync desync corner case due to taking frames from previous response.
- Fixed PIE crash when recording audio at the same time of starting a new connection.
- Fixed minimum buffer duration updated to 0.2 seconds.
- Enhanced GetTestCharacterID to trim whitespace and support command line overrides.

▸ Assets 8

Feb 19
github-actions
4.0.0-beta.11
00040dc
Compare

**Convai Unreal Engine SDK Version 4.0.0-beta.11**

- Improved compatibility with CC5 Reallusion characters.

▸ Assets 8

Feb 11
github-actions
4.0.0-beta.10
b4b94de
Compare

**Convai Unreal Engine SDK Version 4.0.0-beta.10**

- Fixed DLL loading issue
- Fixed crash in the Convai Dashboard
- Added support for Blueprint-only projects

▸ Assets 8

https://github.com/Conv-AI/Convai-UnrealEngine-SDK-V4/releases

# 04. Future Implementations

*Improvements for expanding educational scope and realism*

# Improvements roadmap

## Manual Shock Logic

Expanding the AED protocol to include manual scene safety confirmation and shock delivery.

## CPR Mechanics

Introducing chest compression depth and tempo feedback using Meta Quest hand-tracking.

## Emergency Dispatch

Implementing dynamic conversation trees for realistic dialogue with a 112 operator.

# 05. Demo Time

*Explore the repository and the project documentation below:*

GitHub repository

https://github.com/MN-UniCode/UE_VR_BLSD

Project website

https://mn-unicode.github.io/UE_VR_BLSD/